

RESEARCH INTERESTS

Visual world models, self-supervised visual learning, 3D vision, vision-language and vision-language-action models, and diagnostic evaluation. I study how visual systems can learn predictive representations from time, geometry, and action, and how to evaluate whether those representations support reasoning and interaction beyond benchmark shortcuts.

EDUCATION

4/2025 – 3/2027 (expected) **M.Eng. in Computer Science** University of Tsukuba, Japan
Advisor: [Prof. Yutaka Satoh](#). GPA: 4.27/4.30.

4/2021 – 3/2025 **B.Eng. in Applied Mechanics and Aerospace Engineering** Waseda University, Japan
Advisor: [Prof. Kiyoshi Saito](#). Minor in Computer Science. Undergraduate research on reinforcement learning for energy-system optimization.

SELECTED PUBLICATIONS AND MANUSCRIPTS

1. **Kohsuke Ide**, Ryosuke Yamada, Yue Qiu, Yoshihiro Fukuhara, Hirokatsu Kataoka, Yuki M. Asano, and Yutaka Satoh. ***Beyond Downstream Scores: Controlled Diagnostics for Point-Cloud Self-Supervised Learning Evaluation.*** Conference on Neural Information Processing Systems (**NeurIPS**), 2026 (Evaluations and Datasets Track). **Spotlight (top 1.28%; 48 oral/spotlight papers out of 3,757 submissions).**
2. **Kohsuke Ide**, Ryosuke Yamada, Yue Qiu, Xianzheng Ma, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh. ***Beyond Single Object: Learning 3D Relations with Large Language Models.*** IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (**CVPR Findings**), 2026. [project](#) | [paper](#) | [code](#)
3. **Kohsuke Ide**, Ryosuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh. ***Seeing Red, Thinking Bad: Color Bias in Vision Language Models.*** International Conference on Pattern Recognition (**ICPR**), 2026. [project](#) | [paper](#) | [code](#)
4. **Kohsuke Ide**. ***Consistency Is Not Identification: What Reuse Reveals About Latent Actions.*** Under review at the International Conference on Learning Representations (**ICLR**), 2027.
5. **Kohsuke Ide**, Hiroki Nishizawa, Ryosuke Yamada, Yue Qiu, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh. ***Success Rates Are Not Model Properties: Controlled Diagnostics for VLA Evaluation.*** Under review at the International Conference on Learning Representations (**ICLR**), 2027.
6. Hirokatsu Kataoka^{*}, **Kohsuke Ide**^{*}, Yoshihiro Fukuhara, Shumpei Takezaki, Nakamasa Inoue, Yuki M. Asano, and Christian Rupprecht. ***What Have Vision Models Learned in the Deep Learning Era?*** Under review at the International Conference on Learning Representations (**ICLR**), 2027. ^{*}Equal contribution.
7. **Kohsuke Ide**, Ryosuke Yamada, Yue Qiu, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh. ***Structure First, Appearance Later: Label-Efficient Pretraining for Neural Radiance Fields.*** Under review at the International Conference on 3D Vision (**3DV**) 2027.
8. Ryosuke Yamada, **Kohsuke Ide**, Yoshihiro Fukuhara, Hirokatsu Kataoka, Gilles Puy, Andrei Bursuc, and Yuki M. Asano. ***3D sans 3D Scans: Scalable Pre-training from Video-Generated Point Clouds.*** IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026. [project](#) | [paper](#) | [code](#)
9. Hirokatsu Kataoka, Yoshihiro Fukuhara, Yonglong Tian, Shangzhe Wu, Oishi Deb, Ryosuke Yamada, Christian Rupprecht, Jianyuan Wang, **Kohsuke Ide**, Koichi Namekata, Xianzheng Ma, Yiming Chen, Robert Geirhos, Aditi Raghunathan, Yuki M. Asano, Deva Ramanan, David Fouhey, Andrew J. Davison, Yilun Du, Jiajun Wu, and Zhuang Liu. ***Visual General Intelligence: A White Paper.*** Preprint, 2026. [paper](#)
10. **Kohsuke Ide**, Ryosuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh. ***Colors You Can't See, Semantic Biases You Can't Ignore.*** ICCV Workshop on Multimodal Reasoning and General Intelligence (**ICCV Workshop**), 2025.

RESEARCH EXPERIENCE

- 9/2025 – Present **Research Assistant, AIST** Tsukuba, Japan
- Conduct research on visual world models, 3D understanding, video representation learning, vision-language-action models, and diagnostic evaluation.
 - Lead projects from problem formulation and implementation through experiments and writing; this work has produced first-author papers at **CVPR Findings 2026** and **ICPR 2026**.
 - Collaborate with Prof. Yuki M. Asano on video representation learning, and with researchers at Oxford's Visual Geometry Group on 3D and video understanding.
- 1/2024 – 8/2025 **Research Intern, AIST** Tsukuba, Japan
- Developed diagnostic evaluations for shortcut learning, language leakage, and visual biases in 3D and vision-language models.
 - Built multi-object 3D reasoning benchmarks that require differences between scenes rather than language-only cues.
 - Designed and implemented a video-to-point-cloud data-generation pipeline for scalable 3D pre-training.
- 8/2024 – 3/2025 **Research Intern, Preferred Networks** Tokyo, Japan
- Developed and improved [3D reconstruction and free-viewpoint video](#) pipelines.
 - Optimized rendering and model components for efficient experimentation.
- 9/2021 – 3/2024 **Machine Learning Engineer Intern, Lightblue Technology** Tokyo, Japan
- Developed and deployed an object-tracking system for a work-efficiency monitoring application using Docker and AWS EC2.
 - Improved human-action-recognition accuracy by 20% through feature engineering and [Optuna](#)-based hyperparameter search.
- 9/2023 – 10/2023 **MLOps Engineer Intern, M3** Tokyo, Japan
- Built [RecoverOOM](#), an open-source, event-driven service for automatically recovering GKE CronJobs after out-of-memory failures.
 - Implemented OOM detection through Cloud Logging and Pub/Sub, automatic memory-limit updates, and immediate job re-runs.

RESEARCH VISITS & INTERNATIONAL COLLABORATION

- 19–31 Jan 2026 **Research Visit** University of Oxford, UK
Research meetings with members of the Visual Geometry Group and Active Vision Laboratory.
- 22–30 Apr 2026 **ASPIRE Computer Vision Workshop and Research Visit** University of Oxford, UK
Presented research and discussed follow-up work with researchers at Oxford VGG.

INVITED TALKS

- **Seeing What Matters, Learning How to Find Out: Toward Visual General Intelligence.**
CMU Video Model Journal Club (online), 18 Sep 2026 (PT). [recording](#)

ACADEMIC SERVICE

- **Workshop Organizer**, [Visual General Intelligence: Vision Research Toward the AGI Era](#), CVPR 2026.

HONORS & AWARDS

- **MIRU 2025 Interactive Presentation Award** (Jul 2025)
Selected as one of **11** award recipients from **601** interactive presentations (**top 1.8%**) for "Can 3D Large Language Models Count to Three?"
- **WINC Hackathon Winner** (2023)
Built a ResNet-based push-up counter optimized with TensorRT for Jetson Nano.

TECHNICAL SKILLS & LANGUAGES

- **Research:** Computer vision, visual world models, self-supervised learning, 3D reconstruction, vision-language and vision-language-action models, diagnostic evaluation.
- **Programming and systems:** Python, C++, Go, PyTorch, CUDA, TensorRT, OpenCV, Git, Docker, Kubernetes, Linux, AWS, and GCP.
- **Languages:** Japanese (native); English (fluent; TOEFL iBT 118/120, TOEIC 990/990).